Charge

Tuesday, September 16, 2025 7:49 PM

Conservation of Charge

When we rub our glass rod against silk, a positive charge is produced on glass and a corresponding negative charge is produced on silk. This proves that rubbing doesn't create any charge but transferred from one place to another. This is known as conservation of charge.

Examples of Charge Conservation

Pair Production

When a high energy ray photon strike the heavy nucleus, the energy of the photon is converted into
an electron-positron pair. This process is called pair production. This process is written as;
$\gamma \rightarrow E^{-} + E^{+}$

Pair Annihilation

When an electron of charge –e comes close to a positron of charge +e , they disappear and their rest
mass energy is converted into radiant energy which appear in the form of two oppositely directed -ray
photons.
$E^{-} + E^{+} \rightarrow \gamma + \gamma$

Quantization of Charge

When a physical quantity such as charge exist only in discrete packets rather than in continuously variable amount, We say that the quantity is quantized.

Charge

Any charge that can be measured and observed can be written as:
$q = ne$
Where, n is number of electrons and e is the amount of charge on single electron which is 1.6×10^{-19}C.

Conductors and Insulators

Saturday, November 1, 2025 3:49 PM

Insulators

The materials through which the charges don't flow are called insulators. Glass, chemically pure water, and plastics are common examples of insulators. If the charge is placed on an insulator, the
charges will stay where they are placed.

Conductors

The materials through which charge can flow easily are called conductors. Metals in general, tap
water and the human body are common examples of conductors. The copper rod cannot be charged
because any charges placed on it easily flow through the rod, through your body (which is also a conductor), and to the ground. The insulating handle, however, blocks the flow and allows charge to
build up on the copper.

Coulomb's Law

Tuesday, September 16, 2025 8:08 PM

Coulomb's Law

Electric force is directly proportional to the product of magnitude of charges and inversely proportional to the square of distance between their centers.

$F \propto q_1 q_2$

$F \propto \frac{1}{r^2}$

So,

$F \propto \frac{q_1 q_2}{r^2}$

For ending proportionality, We take constant k,

$F = k\frac{q_1 q_2}{r^2}$

Where, k is called coulomb's constant. Its value depends upon the system of units and medium between the charges. For free space and in system international 'k' is expressed as:

$k = \frac{1}{4\pi\varepsilon_o}$

Where, ε_o is permitivity of free space and its value in SI unit is:

$\varepsilon_o = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$

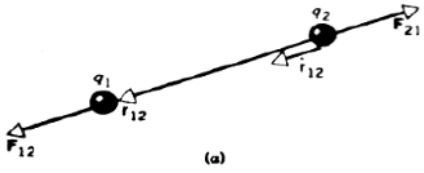
Vector Form of Coulomb's Law

$\overrightarrow{F_{21}} = \frac{1}{4\pi\varepsilon_o} \cdot \frac{q_1 q_2}{r^2} \widehat{r_{21}}$

$\overrightarrow{F_{12}} = \frac{1}{4\pi\varepsilon_o} \cdot \frac{q_1 q_2}{r^2} \widehat{r_{12}}$

$\widehat{r_{12}} = -\widehat{r_{21}}$

Now, we can write, $\overrightarrow{F_{21}} = \frac{1}{4\pi\varepsilon_o} \cdot \frac{q_1 q_2}{r^2}(-\widehat{r_{12}})$

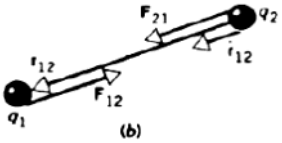


$\overrightarrow{F_{21}} = -\frac{1}{4\pi\varepsilon_o} \cdot \frac{q_1 q_2}{r^2}(\widehat{r_{12}})$

Also,

$\overrightarrow{F_{12}} = \frac{1}{4\pi\varepsilon_o} \cdot \frac{q_1 q_2}{r^2}(-\widehat{r_{21}})$

$\overrightarrow{F_{12}} = -\frac{1}{4\pi\varepsilon_o} \cdot \frac{q_1 q_2}{r^2}(\widehat{r_{21}})$



Significance of Coulomb's Law

- **Explains Atomic Structure**
  On a microscopic level, Coulomb's Law explains what holds an atom together. It's the attractive force between the positive protons in the nucleus and the negative electrons orbiting it.
- **Governs Chemical Bonding**
  The forces that bind atoms together to form molecules (like ionic and covalent bonds) are fundamentally governed by the Coulomb forces between the nuclei and electrons of the different atoms.

Electric Field

Saturday, November 1, 2025 3:48 PM

Electric Field

The region or space around the charge within which its electrical force can be felt by other charges is called Electric field.

Electric field intensity is a single vector quantity which determines the field strength and its direction at a point.

It is denoted by E which is given as:

$$\vec{E} = \frac{\vec{f}}{q_0}$$

Electric Field due to Point Charge

$$According\ to\ Coulomb's\ Law, \vec{F} = \frac{qq_o}{4\pi \in_{0.} r^2}$$

As we know that, $\overrightarrow{E} = \frac{\vec{f}}{q_0}$

So, $\overrightarrow{E} = \frac{qq_o}{4\pi \in_{0.} r^2 . q_o}$

$$\vec{E} = \frac{q}{4\pi \in_{0.} r^2}$$

Electric Lines of Force

It is a path followed by an isolated unit positive charge in an electric field. It may be straight or curved. The tangent to it at any point gives direction of electric field intensity at that point. The concept of electric lines of force was given by Michael Faraday. Electric lines provide information about strength and electric field. The field is stronger near charge where field lines are closer to each other.

These lines must begin on a positive and terminate on a negative terminal due to a point charge. Consider a point charge q placed at point O in order to find electric field intensity at that point P, a test charge q_0

is placed at point P in the electric field of point charge q at a point p in the electric field of point charge q at a distance r from point O.

~~Electric Field due to Dipole~~

Wednesday, October 1, 2025 8:37 PM

Electric Dipole

2 charges of opposite signs and equal magnitude separated by a small distance form an electric dipole.

Dipole Moment

The Product of magnitude of charge and displacement between the charges. It is written as P.

Net Electric Field at a Point

In order to find the electric field we find it individually on x-axis and y-axis and add their squares with a square root applied to their result:

$E=\sqrt{e_x^2 + e_y^2}$

Electric field at X-axis:

$\overrightarrow{E_x} = \overrightarrow{E_{+x}} + \overrightarrow{E_{-x}}$

Since, both are in the opposite direction

$\overrightarrow{E_x} = \overrightarrow{E_{+x}} - \overrightarrow{E_{-x}}$

Since both are of same magnitude they cancel each other

$\overrightarrow{E_x} = 0$

Electric field at Y-axis:

Since both vectors in Y-axis are in same direction.

$\overrightarrow{E_y} = \overrightarrow{E_{+y}} + \overrightarrow{E_{-y}}$

Since both charges have same magnitude and direction we can write both as a positive charge

$\overrightarrow{E_y} = \overrightarrow{E_{+y}} + \overrightarrow{E_{+y}}$

$\overrightarrow{E_y} = 2\overrightarrow{E_{+y}}$

Now, as we know that we write sin because of y-axis

$E_y = 2E\ sin\theta$

We studied earlier that:

$E = \frac{kq}{r^2}$

So, putting its value in the equation,

$E_y = 2\frac{kq}{r^2} sin\theta$

$\sin\theta$:

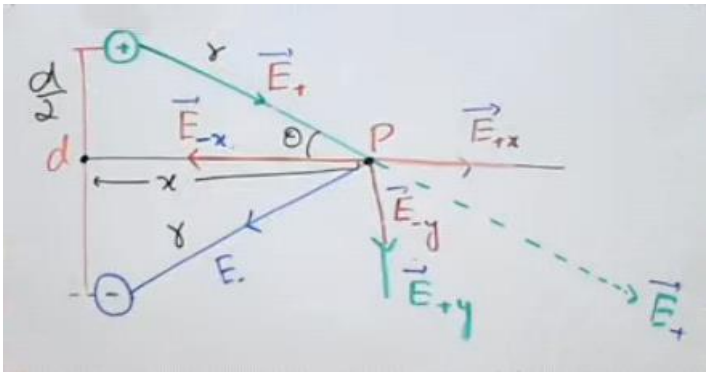
If we observe the figure we can see that there is triangle forming in the figure at point P. Now, we know that:

$sin\theta = \frac{p}{h}$

We can see that in the triangle the perpendicular is the half of distance d between both charges so it is written as d/2. whereas, the hypotenuse is x.

$sin\theta = \frac{\frac{d}{2}}{r}$

$sin\theta = \frac{d}{2r}$



Putting its value in the equation:

$E_y = 2\frac{kq}{r^2}\frac{d}{2r}$

$E_y = \frac{kqd}{r^3}$

As we studied earlier, the Product of magnitude of charge and displacement between the charge (qd) is written as P

$$E_y = \frac{kp}{r^3}$$

Net Force

$$E=\sqrt{(0)^2+\left(\frac{kp}{r^3}\right)^2}$$

$$E = \frac{kp}{r^3}$$

Electric Field due to Ring of Charges

Sunday, October 5, 2025 2:05 PM

$\lambda = \frac{dq}{ds}$

$dq = \lambda ds$

$\int dq = \lambda \int ds$

$q = \lambda \int ds$

Calculation for Net Electric Field:

$dE = \sqrt{dE_x^2 + dE_y^2}$

X-component:

$dE_x = dE_{Lx} - dE_{Mx}$

Because both are in opposite direction and of same magnitude

$dE_L = -dE_M$

$dE_L = \frac{kdq}{r^2}$

$dE_x = 0$

y-component:

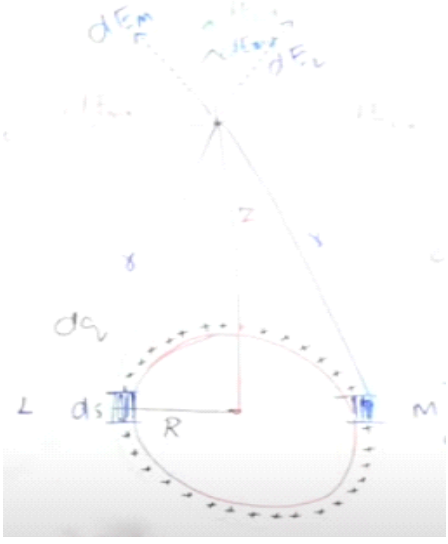
$dE_y = dE_{Ly} - dE_{My}$

Because both are in same direction and equal in magnitude, they add up to give the final Vaule of dE_y:

$dE_y = dE_{Ly}$

$dE_y = dE_{Ly} \sin\theta$ (We write sin with Y and cos with x-components)

Electric Field due to a patch



Putting the values

$dE = \sqrt{0^{2+(dE_{Ly}\sin\theta)^2}}$

$dE = dE_L \sin\theta$

$dE = \frac{kdq}{r^2}\sin\theta$

Considering a triangle in figure:

$\sin\theta = \frac{perpendicular}{hypotenuse}$

$\sin\theta = \frac{z}{r}$

$dE = \frac{kdq}{r^2}.\frac{z}{r}$

$dE = \frac{kdqz}{r^3}$

By Pythagorean theorem:

$hypotenuse^2 = base^2 + perpendicular^2$

$r^2 = R^2 + z^2$

$r = \sqrt{R^2 + z^2}$

$r^3 = (R^2 + z^2)^{3/2}$

$dE = \frac{kdqz}{(R^2 + z^2)^{3/2}}$

Electric Field due to Ring of Charge

$\int dE = \int \frac{kdqz}{(R^2 + z^2)^{3/2}}$

as we saw above, dq= $\lambda \int ds$

$$E = \frac{kz\lambda}{(R^2 + z^2)^{3/2}} \int ds$$

$$E = \frac{kz\lambda \int ds}{(R^2 + z^2)^{3/2}}$$

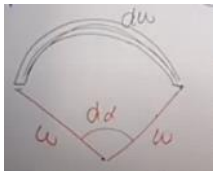
As we know that, $q = \lambda \int ds$

$$E = \frac{q.\, kz}{(R^2 + z^2)^{3/2}}$$

Electric Field Due to a Disk of Charge

Wednesday, October 8, 2025 7:45 PM

Consider a circular disk of uniform surface charge density as shown in the figure below. We want to find out the electric field intensity at point 'P' which is at the distance 'z' from the plane of disk.
Consider a small element of the disk in the ring shape of radius 'ω' and width 'dω'. If 'dq' is the charge on this element of ring having surface area da , then



$$\sigma = \frac{dq}{da}$$

$$\int dq = \sigma \int da$$

$$q = \sigma \int da$$

Thus, $q = \sigma . 2\pi\omega . d\omega$

Electric Field due to selected patches

$$dE_L = \frac{kdq}{r^2}$$

$$dE_M = \frac{kdq}{r^2}$$

$$dE_L = dE_M$$

$$dE = \sqrt{dE_x^2 + dE_y^2}$$

Electric Field at X-Axis:

$$dE_x = dE_{Lx+} dE_{Mx}$$

Since, both are in opposite direction and have same magnitude

$$dE_x = dE_{Lx-} dE_{Mx}$$

$$dE_x = 0$$

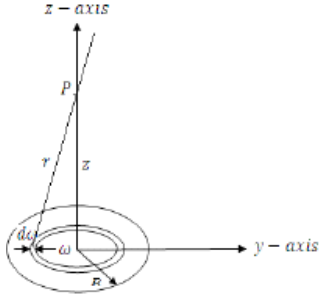
Electric Field at Y-Axis:

$$dE_y = dE_{Ly+} dE_{My}$$

Since, both are in same direction and have same magnitude

$$dE_{Ly} = dE_{My}$$

So, $dE_y = dE_{L \sin\theta}$



Electric Field due to ring of charge

$$dE = \sqrt{0 + (dE_{L \sin\theta})^2}$$

$$dE = dE_{L \sin\theta}$$

As we know,

$$dE_L = \frac{kdq}{r^2}$$

$$dE_L = \frac{k\sigma . 2\pi\omega . d\omega}{r^2}$$

Now, if we consider a triangle inside the disk and apply Pythagorean theorem

$$r^2 = \omega^2 + z^2$$

$$r = \sqrt{\omega^2 + z^2}$$

$$r^3 = (\omega^2 + z^2)^{3/2}$$

Now, if we check for sin θ

$$sin\theta = \frac{z}{r}$$

$$dE = \frac{k . \sigma . 2\pi\omega . d\omega}{r^2} . \frac{z}{r}$$

$$dE = \frac{k . \sigma . 2\pi\omega . d\omega . z}{r^3}$$

$$dE = \frac{k . \sigma . 2\pi\omega . d\omega . z}{(\omega^2 + z^2)^{3/2}}$$

Electric Field due to Disk of Charge:

Since the value of radius is 0 at centre and R at the boundary

$$\int dE = k\sigma z\lambda\pi \int_0^R \frac{2\omega d\omega}{(\omega^2 + z^2)^{3/2}}$$

$$E = k\sigma z\pi \int_0^R 2\omega d\omega . \left(\omega^2 + z^2\right)^{-3/2}$$

$$E = k\sigma z\pi \frac{\left(\omega^2 + z^2\right)^{-1/2}}{\frac{-1}{2}} \int_0^R$$

$$E = -2k\sigma z\pi \frac{1}{(z^2 + \omega^2)^{1/2}} \int_0^R$$

Expanding k and applying limits

$$E = \frac{-2\sigma z\pi}{4\pi\epsilon_o}\left(\frac{1}{(z^2 + R^2)^{1/2}} - \frac{1}{(z^2 + 0)^{1/2}}\right)$$

$$E = \frac{-2\sigma z\pi}{4\pi\epsilon_o}\left(\frac{1}{(z^2 + R^2)^{1/2}} - \frac{1}{(z^2 + 0)^{1/2}}\right)$$

$$E = \frac{\sigma z}{2\epsilon_o}\left(\frac{1}{z} - \frac{1}{(z^2 + R^2)^{1/2}}\right)$$

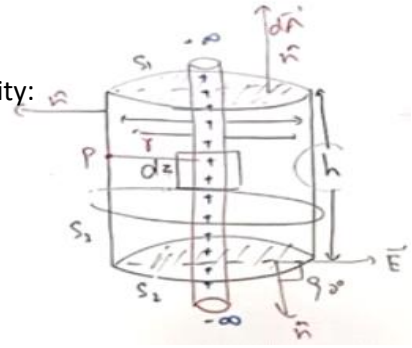
$$E = \frac{\sigma}{2\epsilon_o}\left(\frac{z}{z} - \frac{z}{(z^2 + R^2)^{1/2}}\right)$$

$$E = \frac{\sigma}{2\epsilon_o}\left(1 - \frac{z}{(z^2 + R^2)^{1/2}}\right)$$

~~Electric Field due to Infinite Line of Charges~~

Thursday, October 9, 2025 8:25 PM

We suppose that there is a rod that is positively charged which charges forming a line on which charges are uniformly distributed.
Firstly, we consider a gaussian surface which helps us find the Electric Field. The length of Gaussian surface is written as h.



Charge Density
We can see that charges are distributed in a line so we use linear charge density:

$$\lambda = \frac{d_q}{d_s}$$

$$\int d_q = \lambda \int ds$$

$$q = \lambda h$$

Electric Flux by definition

We choose an arbitrary point p to find electric flux at that point
Now, Electric flux by definition is given as:

$$\varphi = \int \vec{E}.\overrightarrow{da}$$

Now, we simplify dot product so,

$$\varphi = \int Edacos\theta$$

Now, we find the electric flux on three surfaces of cylinder so we draw vector of 3 surfaces

$$\varphi_1 = \int Edacos\theta$$

As we can see that the normal vector and electric field lines are perpendicular

$$\varphi_1 = \int Edacos90$$

As we know that, *cos90=0*

$$\varphi_1 = 0$$

we can see that the normal vector and electric field lines are perpendicular

$$\varphi_2 = 0$$

Now, we can see that in case of third surface, the normal vector and electric field lines are p*arallel*

$$\varphi_3 = \int Edacos0$$

As we know, *cos0=1*

$$\varphi_3 = E\int da$$

Now, we know that the circumference of cylinder is $2\pi r$ and height is h so,

$$\varphi_3 = E(2\pi r)h$$

For total flux:

$$\varphi_e = \varphi_1 + \varphi_2 + \varphi_3$$

$$\varphi_e = 0 + 0 + E(2\pi r)h$$

$$\varphi_e = E(2\pi r)h$$

Electric Flux due to Gauss's Law

As we know that according to Gauss's Law

$$\varphi_e = \frac{1}{\epsilon_o} q$$

Comparing with the equation we found earlier,

$$E(2\pi r)h = \frac{1}{\epsilon_o} q$$

$$E(2\pi r)h = \frac{1}{\epsilon_o} \lambda h$$

$$E = \frac{1}{\epsilon_o.(2\pi r)h} \lambda h$$

$$E = \frac{\lambda}{\epsilon_o.(2\pi r)}$$

$$\vec{E} = \frac{\lambda}{\epsilon_o . (2\pi r)} \hat{r}$$

Electrostatics Numerical

Sunday, October 12, 2025 2:43 PM

1. Coulomb's Law

Two point charges $q_1 = 9 \times 10^{-6} C$ and $q_2 = -3 \times 10^{-6} C$ are placed 0.3 m apart in vacuum.

Find: The force between them (magnitude and nature: attractive or repulsive).

Given:

- $q_1 = 9 \times 10^{-6} C$
- $q_2 = -3 \times 10^{-6} C$
- r=0.3 m
- $k = 9 \times 10^{9} Nm^2 C^{-2}$

Formula:

$$F = k \frac{q_1 q_2}{r^2}$$

Calculation:

$$F = 9 \times 10^{9} \frac{9 \times 10^{-6} . -3 \times 10^{-6} C}{(0.3)^2}$$

$F = 2.7$ N

Nature is attractive because of opposite signs

2. Comparison of Electrostatic vs Gravitational Force

Calculate the ratio of electrostatic force to gravitational force between an electron and proton separated by 5.3 x 10^{-11} m.

Given:

- $r = 5.3 \times 10^{-11} m$
- $e = 1.6 \times 10^{-19} C$
- $m_e = 9.11 \times 10^{-31} kg$
- $m_p = 1.67 \times 10^{-27} kg$
- $k = 9 \times 10^{9} Nm^2 C^{-2}$
- $G = 6.67 \times 10^{-11}$

Formula:

$$Ratio = \frac{F_e}{F_g} = \frac{ke^2/r^2}{Gm_1 m_2/r^2} = \frac{ke^2}{Gm_1 m_2}$$

Calculation:

$$\frac{F_e}{F_g} = \frac{9 \times 10^{9} . (1.6 \times 10^{-19})^2}{6.67 \times 10^{-11} . 9.11 \times 10^{-31} . 1.67 \times 10^{-27}}$$

$$\frac{F_e}{F_g} = 2.3 \times 10^{39}$$

3. Superposition Principle

Three charges (+2 , μ C), (-2 , μ C), and (+1 , μ C) are placed at equal separations of 0.4 m on a straight line.

Find: The net force on the (+1 , μ C) charge.

Given:

- $q_1 = 2 \times 10^{-6}C$
- $q_2 = -2 \times 10^{-6}C$
- $q_3 = 1 \times 10^{-6}C$
- $k = 9 \times 10^{9}Nm^2C^{-2}$

Formula:

$F = k\frac{q_1 q_2}{r^2}$

Calculation:

$F_1 = 9 \times 10^{9}\frac{2 \times 10^{-6}.1 \times 10^{-6}C}{(0.8)^2}$

$F_1 = 0.028N$

$F_2 = 9 \times 10^{9}\frac{-2 \times 10^{-6}.1 \times 10^{-6}C}{(0.4)^2}$

$F_2 = 0.112N$

$Net\ Force = F = F_2 - F_1$

$F = 0.112 - 0.028$

$F = 0.084N$

4. Electric Field Due to Point Charge

A point charge (+5 , m C) is placed at the origin.

Find: The electric field intensity at a point on the x-axis, 2 m away.

Given:

- $q = 5 \times 10^{-6}C$
- $r = 2m$
- $k = 9 \times 10^{9}Nm^2C^{-2}$

Formula:

$E = k\frac{q}{r^2}$

Calculation:

$E = 9 \times 10^{9}\frac{5 \times 10^{-6}}{(2)^2}$

$E = 9 \times 10^{9}\frac{5 \times 10^{-6}}{(2)^2}$

$E = 1.125 \times 10^{4}$

5. Dipole Moment

Find the dipole moment of an electron and proton separated by a distance of 5 x 10^{-11} m.

Given:

- $q = 1.6 \times 10^{-19}C$
- $d = 5 \times 10^{-11}m$

Formula:

$P = qd$

Calculation:

$P = 1.6 \times 10^{-19}.5 \times 10^{-11}$

$P = 8 \times 10^{-30}C\ m$

6. Electric Field Due to Dipole

A dipole of moment (3.56 x10^{-29} , C m) is placed along its bisector axis at a distance (25 , nm).

Find: The electric field intensity at that point.

Given:

- $k = 9 \times 10^{9} Nm^{2}C^{-2}$
- $p = 3.56 \times 10^{-29} C\ m$
- $r = 25nm = 25 \times 10^{-9} m$

Formula:

$$E = \frac{kp}{r^{3}}$$

Calculation:

$$E = \frac{9 \times 10^{9}.\ 3.56 \times 10^{-29}}{(25 \times 10^{-9})^{3}}$$

$$E = 2.04 \times 10^{4} NC^{-1}$$

7. Work and Potential

A charge of 2μC is moved from a point at potential 20 V to a point at potential 80 V.

Find: The work done by the electric field.

Given:

- $q = 2 \times 10^{-6} C$
- $\Delta V = 80 - 20 = 60V$

Formula:

$W = q\Delta V$

Calculation:

$W=2 \times 10^{-6}.\ 60$

$W=1.2 \times 10^{-4} J$

8. Electric Field Intensity

A point charge of 2μC is placed directly at a distance of 2cm from point P. Calculate Electric Field Intensity at point P.

Given:

- $q= 2\mu C=2 \times 10^{-6} C$
- $r= 2cm=0.02m$

Formula:

$$E = k\frac{q}{r^{2}}$$

Calculation:

$$E = \frac{9 \times 10^{9}\ .\ 2 \times 10^{-6}}{(0.02)^{2}}$$

$$E = 4.5 \times 10^{7} NC^{-1}$$

9. Gauss's Law – Point Charge

A point charge of (1.84 x10^{-6} C) is placed at the center of a cube of side 0.55 m.

Find: The total electric flux through the cube.

Given:

- $q = 1.84 \times 10^{-6} C$

- $\epsilon_o = 8.85 \times 10^{-12}$

Formula:

$\varphi_E = \frac{q}{\epsilon_o}$

Calculation:

$\varphi_E = \frac{1.84 \times 10^{-6}}{8.85 \times 10^{-12}}$

$\varphi_E = 2.08 \times 10^5 Nm^2C^{-1}$

10. Gauss's Law – Charged Conducting Sphere

A conducting sphere of radius 0.5 m has a surface charge density of (2 x10^{-6} ,$\mathbf{Cm^{-2}}$).

Find: The total electric flux through the sphere.

Given:

- $\sigma = 2.0 \times 10^{-6} Cm^{-2}$
- $r = 0.50m$
- $\epsilon_o = 8.85 \times 10^{-12}$

Formula:

$q = \sigma(4\pi r^2)$

$\varphi_E = \frac{q}{\epsilon_o}$

Solution:

$A = (4)(3.14)(0.50)^2$

$A = 3.14m^2$

$q = 2.0 \times 10^{-6}.3.14$

$q = 6.28 \times 10^{-6} C$

$\varphi_E = \frac{6.28 \times 10^{-6}}{8.85 \times 10^{-12}}$

$\varphi_E = 7.10 \times 10^5 Nm^2C^{-1}$

Dipole in an Electric Field

Monday, October 13, 2025 10:41 AM

Electric Dipole

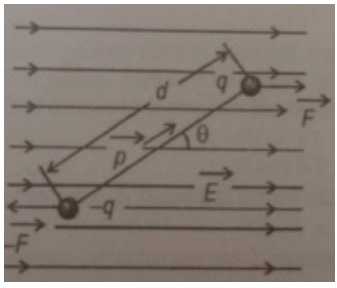
2 charges of opposite signs and equal magnitude separated by a small distance form an electric dipole.

Dipole Moment

The Product of magnitude of charge and displacement between the charges. It is written as p.

Torque on Dipole in an Electric Field

When we place a dipole in an electric field, the force on positive charge will be in one direction and the force on negative charge will be in other direction. To account for the effect of these forces, we introduce a dipole moment vector p. The vector p has magnitude qd and direction along the line joining from positive to negative charge.



$\tau = F\frac{d}{2}\sin\theta + F\frac{d}{2}\sin\theta$

$\tau = Fd\sin\theta$

As we know that, $F = qE$

$\tau = qEd\sin\theta$

$\tau = qdE\sin\theta$

$\tau = pE\sin\theta$

In vector form, $\vec{\tau} = \vec{p} \times \overrightarrow{E}$

Energy of Dipole in an Electric Field

The work done by external electric field to turn the dipole from initial angle θ_0 to θ is given as:

$$W = \int fd$$

Since, f is given as torque and d is given as $d\theta$

$$W = \int_{\theta_0}^{\theta} \vec{\tau}.\overrightarrow{d\theta}$$

Since, $\vec{\tau}$ and $\overrightarrow{d\theta}$ are in opposite direction,

$$W = \int_{\theta_0}^{\theta} \tau.\, d\theta\, cos180$$

As we know that *cos 180=-1 and* $\tau = pE\sin\theta$

$$W = -\int_{\theta_0}^{\theta} pE\sin\theta\,.\,d\theta$$

$$W = -pE\int_{\theta_0}^{\theta} \sin\theta\,.\,d\theta$$

$$W = -pE\int_{\frac{\pi}{2}}^{\theta} \sin\theta\,.\,d\theta$$

$$W = pE\left(\cos\theta - \cos\frac{\pi}{2}\right)$$

$$W = pE(\cos\theta)$$

Since work done by agent that produces external field is equal to negative of change in potential energy U of the system

$U = -W$

$U = -pE(\cos\theta)$

$U = -\vec{p}.\vec{E}$

This is the expression of potential energy of the dipole in a uniform electric field.

Electric Flux

Monday, October 13, 2025 11:28 AM

Electric Flux

Electric flux means the lines of force passing through a given area given the area is perpendicular to lines of force. For a uniform electric field, electric flux $\emptyset_e$ through an area is defined as:

The product of area and component of electric field normal to area

$\emptyset_e = EA \cos\theta$

Where θ is the angle between E and positive normal to area A. In vector form,

$\emptyset_e = \vec{E}.\vec{A}$

For a non-uniform electric field $\vec{E}$ varies from point to point, we divide the surface into small elements each of area d$\vec{a}$

$d\emptyset_e = \vec{E}.d\vec{A}$

The total electric flux over whole area is given as:

$$\emptyset_e = \sum \vec{E}.d\vec{A}$$

If $d\vec{A}$ approaches 0, then summation can be replaced by integral,

$$\emptyset_e = \int_S \vec{E}.d\vec{A}$$

We can see that electric flux through a vector area $d\vec{A}$ is given by scalar product of Electric field vector $\vec{E}\ and\ vector\ d\vec{A}$. The unit of electirc flux is Nm^2C^{-1}.

Gauss's Law

Tuesday, October 14, 2025 12:08 PM

Gauss's Law

The total normal electric flux which diverging out of a closed surface is equal to $\frac{1}{\varepsilon_o}$ times the total charge enclosed by the surface.

Mathematically, $\emptyset_e = \frac{1}{\varepsilon_o} Q$

Where, Q is charge enclosed within a surface

Consider a charge q inside a closed surface S. Take an area element $\overrightarrow{da}$ which makes an angle θ with with electric field intensity due to point charge q at point P, which is situated at distance r from origin. The electric field intensity at point point p due to charge q is,

$\vec{E} = \frac{1}{4\pi\varepsilon_o} . \frac{q}{r^2} \hat{r}$

The outward normal flux through $\overrightarrow{da}$ is given by:

$d\emptyset_e = \vec{E} . d\vec{A} = \frac{1}{4\pi\varepsilon_o} . \frac{q}{r^2} \hat{r} . da$

Total Electric flux through whole surface is,

$$\emptyset_e = \int_s \frac{q}{4\pi\varepsilon_o} . \frac{\hat{r} . da}{r^2} = \frac{q}{4\pi\varepsilon_o} \int_s \frac{\hat{r} . da}{r^2}$$

Now, $\int_s \frac{\hat{r} . da}{r^2} = 4\pi$, so,

$\emptyset_e = \frac{q}{4\pi\varepsilon_o} . 4\pi$

$\emptyset_e = \frac{q}{\varepsilon_o}$

If there are n point charges, say $q_1, q_2, q_3 \ldots q_n$, then total electric flux through enclosed surface point chages is,

$\emptyset_e = \frac{q_1}{\varepsilon_o} + \frac{q_2}{\varepsilon_o} + \cdots \frac{q_n}{\varepsilon_o}$

$\emptyset_e = \frac{1}{\varepsilon_o} (q_1 + q_2 + \cdots q_n)$

$\emptyset_e = \frac{1}{\varepsilon_o} Q$

Where, $Q = q_1 + q_2 + \cdots q_n$ is the total charge enclosed within surface

Differential form of Gauss's Law

By definition of electric flux,

$$\emptyset_e = \int_s \vec{E} . d\vec{A}$$

According to Gauss's Law,

$\emptyset_e = \frac{1}{\varepsilon_o} Q$

In case of volume charge distribution,

$$q = \int_v \rho \, dV$$

So,

$$\emptyset_e = \frac{1}{\varepsilon_o} \int_v \rho \, dV$$

Comparing eq. above,

$$\int_s \vec{E} . d\vec{A} = \frac{1}{\varepsilon_o} \int_v \rho \, dV$$

Applying divergence theorem on left side of equation,

$$\int_V^{\square} div\vec{E}.\, d\vec{V} = \int_v^{\square} \frac{1}{\varepsilon_o}.\, \rho\, dV$$

Since, above equation is true for any volume element so,

$$div\vec{E} = \frac{\rho}{\varepsilon_o}$$

This is differential form of Gauss's Law

Integral form of Gauss's Law

Differential form of Gauss's Law is given as:

$$div\vec{E} = \frac{\rho}{\varepsilon_o}$$

Integrating over volume, we have:

$$\int_V^{\square} div\vec{E}.\, d\vec{V} = \int_v^{\square} \frac{\rho}{\varepsilon_o} dV$$

Applying divergence theorem on left side of equation,

$$\int_s^{\square} \vec{E}.\, d\vec{A} = \frac{1}{\varepsilon_o} \int_v^{\square} \rho\, dV$$

In case of Volume Charge Distribution,

$$q = \int_v^{\square} \rho\, dV$$

So,

$$\int_s^{\square} \vec{E}.\, d\vec{A} = \frac{1}{\varepsilon_o} Q$$

Applications of Gauss's Law

Tuesday, October 14, 2025 2:42 PM

A charged isolated conductor: Gauss's Law permits us to prove an important theorem for isolated conductor.
The theorem states that "An excess charge placed on an isolated conductor moves entirely to outer surface of conductor.
Proof:
Consider a charged conductor and any closed surface inside it as gaussian surface. It is obvious that under static conditions electric field inside conductor vanishes because if it were not so there will be a movement of charges under the influence of field an d static condition will not exist. Hence, flux $\emptyset_e$ through that surface us 0.
By Gauss's Law,

$$\frac{q}{\varepsilon_o} = 0$$

So, q=0

An isolated conductor with cavity

Consider a conductor with a cavity. Draw a gaussian surface surrounding cavity, close to its wall but inside conducting body.
Because, electric field intensity is zero inside conductor, there can be no flux through conductor. i.e.

$$\varepsilon_o = 0$$

$$\frac{q}{\varepsilon_o} = 0$$

So, q=0
We conclude that there is no charge on cavity walls, It remains on the outer surface of conductor

Electric field near a charged conducting sheet:

Consider a charged conductor of certain thickness. Suppose, its upper surface has charge density σ.
We have to calculate the electric field intensity at point P just outside the conductor. Imagine a cylinder with bottom in th e the conductor and top through point P. Let area of top is A. There will be no flux in downward direction because charge inside co nductor is 0. Also, no flux will pass through curved surface of cylinder because E and A are perpendicular. Flux will only pass throu gh top of cylinder, which is given as:

$$\varphi_e = \vec{E}.\vec{A} = EA\cos\theta = EA$$

By Gauss's Law,

$$\varphi_e = \frac{Q}{\varepsilon_o}$$

$$\varphi_e = \frac{\sigma A}{\varepsilon_o}$$

Comparing equations,

$$EA = \frac{\sigma A}{\varepsilon_o}$$

$$E = \frac{\sigma}{\varepsilon_o}$$

In vector's form,

$$E = \frac{\sigma}{\varepsilon_o}\hat{r}$$

Electric Field Intensity due to a Spherically Symmetric Charge Distribution

Case I: Point Outside the Sphere (r>R)

- Total charge Q is uniformly distributed over a sphere of radius R.
- Gaussian surface: sphere of radius r>R
- By Gauss's law:

$$\varphi_e = \int_s \vec{E}.\overrightarrow{da}$$

$$\varphi_e = \int_s Edacos0^o$$

$$\varphi_e = \int_s Eda$$

$$\varphi_e = E \int_s da$$

$$\varphi_e = E.4\pi r^2$$

By Gauss's Law, $\emptyset_e = \frac{Q}{\varepsilon_o}$

$$E.4\pi r^2 = \frac{Q}{\varepsilon_o}$$

$$E = \frac{Q}{\varepsilon_o.4\pi r^2}$$

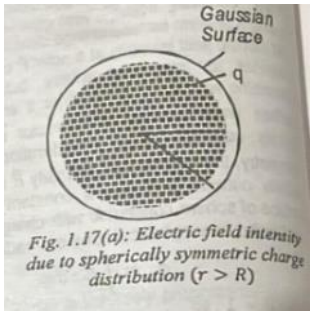


Fig. 1.17(a): Electric field intensity due to spherically symmetric charge distribution (r > R)

Result: Outside the sphere, the field is the same as if all charge were concentrated at the center.

Case II: Point inside the Sphere (r<R)

$\boldsymbol{\rho} = \frac{q}{V}$

$$\rho = \frac{q}{4\pi R^3/_3}$$

$$\rho = \frac{3q}{4\pi R^3}$$

Net charge within gaussian surface of radius r is

$$q^- = \rho\left(\frac{4\pi r^3}{3}\right)$$

$$q^- = \frac{3q}{4\pi R^3}.\left(\frac{4\pi r^3}{3}\right)$$

$$q^- = \frac{qr^3}{R^3}$$

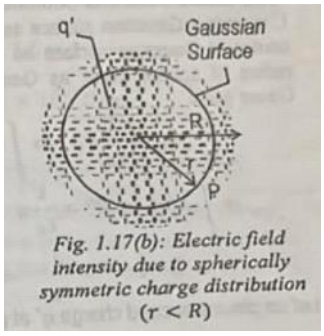


Fig. 1.17(b): Electric field intensity due to spherically symmetric charge distribution (r < R)

Total electric flux through Gaussian surface is

$$\varphi_e = \int_s \vec{E}.\overrightarrow{da}$$

$$\varphi_e = \int_s Edacos0^o$$

$$\varphi_e = \int_s Eda$$

$$\varphi_e = E\int_s da$$

$$\varphi_e = E.4\pi r^2$$

By Gauss's Law

$$\varphi_e = \frac{1}{\epsilon_o}.q^-$$

$$\varphi_e = \frac{1}{\epsilon_o}.\frac{qr^3}{R^3}$$

Comparing the equations

$$E.4\pi r^2 = \frac{1}{\epsilon_o}.\frac{qr^3}{R^3}$$

$$E = \frac{qr}{4\pi\epsilon_o.R^3}$$

Equipotential Surfaces

Wednesday, November 12, 2025 12:28 PM

If all the points on a surface have the same value of electric potential, then, it is known as equipotential surface.
Suppose we have a charge that exerts a field where the other charge can feel the attraction or repulsion. If we want to find the potential difference between 2 points A and B. We consider a hypothetical surface which is an equipotential surface.
$\mathrm{W} = \mathrm{f}.\mathrm{dr}$
As we know, $W = q\Delta V$
And $F = qE$
$q\Delta V = qE.dr$
$q(V_b - V_a) = E.drcos\theta$
Since the angle between E and dr is 90
$q(V_b - V_a) = E.drcos90$
$q(V_b - V_a) = 0$
$V_b - V_a = 0$
$V_b = V_a$

Potential Difference between 2 Points

Friday, October 24, 2025 7:46 PM

Potential Difference between 2 Points

Potential difference between 2 points is defined as "The amount of work done ΔW per unit charge q_o in moving it from one point to another against electric field and by keeping system in equilibrium". Mathematically,

$$\Delta V = \frac{\Delta W}{q_o}$$

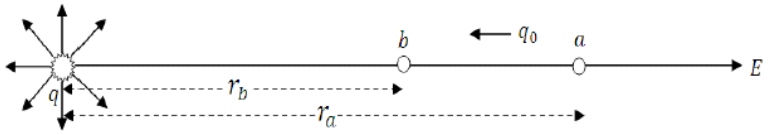
Suppose a unit positive q_o is moved from one point 'a' to the point 'b' in the electric field ' $\vec{E}$ ' of a large positive charge 'q'

The work done in moving this charge from point 'a' to point 'b' is

$$W_{A\to B} = \int_a^b \vec{F}.\overrightarrow{dr}$$

The electric force of magnitude $\vec{F} = -q_o\vec{E}$ must have to be supplied to move test charge against field

$$W_{A\to B} = \int_a^b (-q_o\vec{E}).\overrightarrow{dr}$$



$$W_{A\to B} = -q_o \int_a^b \vec{E}.\overrightarrow{dr}$$

$$\frac{W_{A\to B}}{q_o} = -\int_a^b \vec{E}.\overrightarrow{dr}$$

Now, $\frac{W_{A\to B}}{q_o} = \Delta V = V_b - V_A$

$$V_b - V_A = -\int_a^b \vec{E}.\overrightarrow{dr}$$

Electric Potential due to a Point Charge

Friday, October 24, 2025 8:33 PM

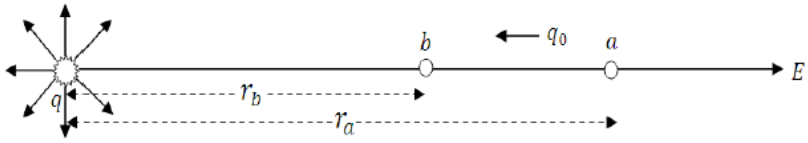
Electric Potential due to a point charge

The potential difference between 2 points is given as:

$$V_B - V_A = -\int_{r_A}^{r_B} \vec{E}.\overrightarrow{dr}$$

Suppose a charge q lies at origin of coordinate system and the radial distances of points A and B are r_A and r_B respectively,
The electric field intensity at a point distance r due to point charge q is,

$$\vec{E} = \frac{1}{4\pi\epsilon_o}.\frac{q}{r^2}\hat{r}$$



$$V_B - V_A = -\int_{A}^{B} \frac{1}{4\pi\epsilon_o}.\frac{q}{r^2}\hat{r}.\overrightarrow{dr}$$

$$V_B - V_A = -\frac{q}{4\pi\epsilon_o}\int_{A}^{B} \frac{\hat{r}.\overrightarrow{dr}}{r^2}$$

$\vec{E}$ is radially directed outward therefore,

$$\hat{r}.\overrightarrow{dr} = (r)(dr)cos0 = (1)dr(1) = dr$$

$$V_B - V_A = -\frac{q}{4\pi\epsilon_o}\int_{A}^{B} \frac{dr}{r^2}$$

$$V_B - V_A = -\frac{q}{4\pi\epsilon_o}(-\frac{1}{r})\int_{r_A}^{r_B}$$

$$V_B - V_A = \frac{q}{4\pi\epsilon_o}\left(\frac{1}{r_B} - \frac{1}{r_A}\right)$$

Suppose, A approaches infinity

$$V_B - 0 = \frac{q}{4\pi\epsilon_o}\left(\frac{1}{r_B} - 0\right)$$

$$V_B = \frac{q}{4\pi\epsilon_o}\left(\frac{1}{r_B}\right)$$

In general, electric potential at a point at a distance r is,

$$V = \frac{q}{4\pi\epsilon_o}.\frac{1}{r}$$

Electric Potential due to a Dipole

Saturday, November 1, 2025 2:52 PM

Consider 2 point charges +q and -q of equal magnitude lying distance 'd' apart.
The electric potential at point 'P' is the sum of potential due to '+q' and '-q'. If $V_+$and V_- are the elctric potential due to charge +q and -q. Therefore,

V = $V_+ + V_-$

V = $\frac{1}{4\pi\epsilon_o}\frac{+q}{r_1} + \frac{1}{4\pi\epsilon_o}\frac{-q}{r_2}$

V = $\frac{q}{4\pi\epsilon_o}\left[\frac{+1}{r_1} + \frac{-1}{r_2}\right]$

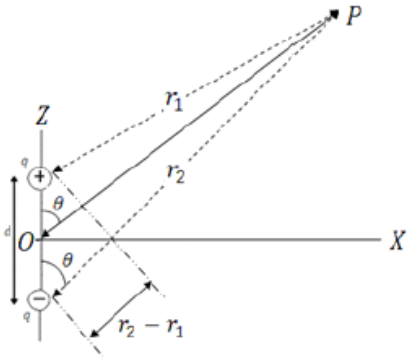
V = $\frac{q}{4\pi\epsilon_o}\left[\frac{r_2 - r_1}{r_1 r_2}\right]$

Normally for a dipole, r>>d

Therefore,

$r_1 r_2 = r^2$

And from fig., $r_2 - r_1 = d\cos\theta$



V = $\frac{q}{4\pi\epsilon_o}\left[\frac{d\cos\theta}{r^2}\right]$

V = $\frac{1}{4\pi\epsilon_o}\frac{qd\cos\theta}{r^2}$

We know that, Dipole Moment, p=qd

V = $\frac{1}{4\pi\epsilon_o}\frac{p\cos\theta}{r^2}$

Multiplying and dividing by r

V = $\frac{1}{4\pi\epsilon_o}\frac{pr\cos\theta}{r^3}$

V = $\frac{1}{4\pi\epsilon_o}\frac{\vec{p}.\vec{r}}{r^3}$

Electric Potential of Continuous Charge Distribution

Friday, October 24, 2025 8:33 PM

Consider a charge distribution with linear charge density λ or a surface charge density σ or a volume charge density ρ. We divide charge distribution into small elements each of chrage dq, where

$dq = \lambda dl$

$dq = \sigma da$

$dq = \rho dv$

Each element dq can be treated as a point charge with a contribution dv to potential given by,

$$dV = \frac{1}{4\pi\epsilon_o}\frac{dq}{r}$$

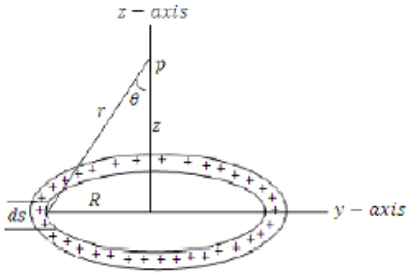
Potential due to whole distribution is,

$$V = \frac{1}{4\pi\epsilon_o}\int\frac{dq}{r}$$

Electric Potential due to a Ring of Charge

Saturday, November 1, 2025 2:13 PM

Consider a ring of positive charge of radius 'R'. We want to find out elecric potential at pont 'P' which is at distance 'z' from plane of ring. As charge is uniformly over it, so it has constant linear charge density λ. For a small length element d_s.



$\lambda = \frac{dq}{ds}$

$dq = \lambda ds$

$q = \lambda \int ds$

The electric potential due to charge dq at a point 'P' is given by:

$dV = \frac{1}{4\pi\epsilon_o}\frac{dq}{r} = \frac{1}{4\pi\epsilon_o}\frac{\lambda ds}{r}$

$V = \int dV = \frac{1}{4\pi\epsilon_o}\int \frac{\lambda ds}{\sqrt{R^2 + z^2}}$

$V = \frac{1}{4\pi\epsilon_o}.\frac{1}{\sqrt{R^2 + z^2}}\int \lambda ds$

As seen above, $q = \lambda \int ds$ so,

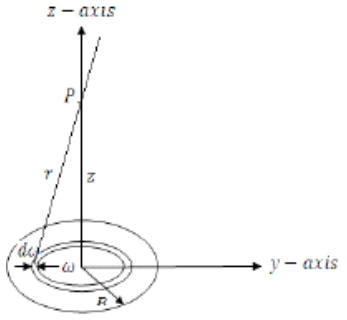
$V = \frac{1}{4\pi\epsilon_o}.\frac{q}{\sqrt{R^2 + z^2}}$

Electric Potential due to a Disk of Charge

Saturday, November 1, 2025 2:25 PM

Consider a circular disk of uniform surface charge density. We want to find out electric potential at point 'P' which is at distance 'z' from plane of disk.

Consider a small element of disk in ring shape of radius 'ω' and width '$d\omega$'. If 'dq' is the charge on this element of ring then,



$$\sigma = \frac{dq}{da}$$

$$dq = \sigma da$$

$$q = \sigma(2\pi\omega.d\omega)$$

We know that electric potential due to a ring of charge which has radius ω is given as,

$$dV = \frac{1}{4\pi\epsilon_o}\frac{dq}{r}$$

$$dV = \frac{1}{4\pi\epsilon_o}\frac{\sigma(2\pi\omega.\,d\omega)}{r}$$

From figure,

$$\text{hypotenuse}^2 = \text{base}^2 + \text{perpendicular}^2$$

$$r^2 = z^2 + \omega^2$$

$$r = \sqrt{z^2 + \omega^2}$$

$$dV = \frac{1}{4\pi\epsilon_o}\frac{\sigma(2\pi\omega.\,d\omega)}{\sqrt{z^2 + \omega^2}}$$

The electric potential at point 'P' due to whole disk is:

$$V = \frac{1}{4\pi\epsilon_o}\int_{\omega=0}^{\omega=R}\frac{\sigma(2\pi\omega.\,d\omega)}{\sqrt{z^2 + \omega^2}}$$

$$V = \frac{\sigma\pi}{4\pi\epsilon_o}\int_{\omega=0}^{\omega=R}\frac{(2\omega.\,d\omega)}{\sqrt{z^2 + \omega^2}}$$

$$V = \frac{\sigma}{4\epsilon_o}\int_{\omega=0}^{\omega=R}(2\omega.\,d\omega).(z^2 + \omega^2)^{\frac{-1}{2}}$$

$$V = \frac{\sigma}{4\epsilon_o}.\frac{(z^2 + \omega^2)^{\frac{-1}{2}+1}}{\frac{1}{2}}\int_0^R$$

$$V = \frac{2\sigma}{4\epsilon_o}.(z^2 + \omega^2)^{\frac{1}{2}}\int_0^R$$

$$V = \frac{\sigma}{2\epsilon_o}.\left((z^2 + R^2)^{\frac{1}{2}} - (z^2)^{\frac{1}{2}}\right)$$

$$V = \frac{\sigma}{2\epsilon_o}.\left(\sqrt{z^2 + R^2} - z\right)$$

Calculating Electric Field from Electric Potential

Wednesday, November 12, 2025 11:42 AM

Consider a test charge q_o is displaced in electric field from point P to point Q. If the potential difference between these points is dV, then work done dW by the electric field is

$dW = -q_o\, dV$

If the test charge is placed in an electric field, then it will experience a force

$\vec{F} = q_o\vec{E}$

The work done in displacing it through displacement is:

$dW = \vec{F}.\overrightarrow{ds}$

$=q_o\vec{E}.\overrightarrow{ds}$

Comparing the equations above:

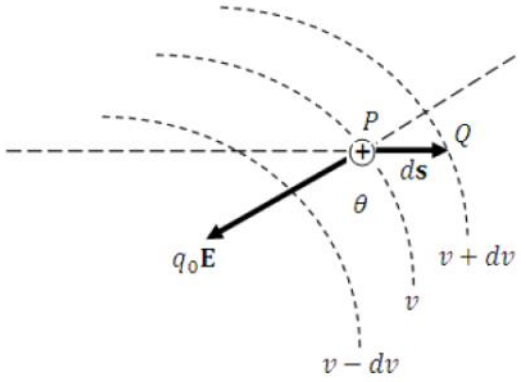
$q_o\vec{E}.\overrightarrow{ds} = -q_o\, dV$

$E.\, ds\cos\theta = dV$

$E\cos\theta = \dfrac{-dV}{ds}$

Considering $E\cos\theta$ as a compont of E along ds

$E_s = \dfrac{-dV}{ds}$



There is only one direction in which the rate of change of electric potential w.r.t position is maximum, which is direction of electric field. Thus:

$E = -(\dfrac{dV}{ds})_{max}$

The maximum value of $\frac{dV}{ds}$ at any point is called the potential graident of electric field.

$E = -\nabla V$

$E = -grad\, V$

Electric Potential due to point charge is:

$V = \dfrac{1}{4\pi\epsilon_o}.\dfrac{q}{r}$

$Now, r = \sqrt{x^2+y^2+z^2}$

$V = \dfrac{1}{4\pi\epsilon_o}.\dfrac{q}{\sqrt{x^2+y^2+z^2}}$

Now, $grad\, V = \left(\dfrac{\partial}{\partial x}\hat{\imath} + \dfrac{\partial}{\partial y}\hat{\jmath} + \dfrac{\partial}{\partial z}\hat{k}\right)V$

$=\dfrac{\partial V}{\partial x}\hat{\imath} + \dfrac{\partial V}{\partial y}\hat{\jmath} + \dfrac{\partial V}{\partial z}\hat{k}$

Consider, $\dfrac{\partial V}{\partial x} = \dfrac{\partial}{\partial x}\left(\dfrac{1}{4\pi\epsilon_o}.\dfrac{q}{\sqrt{x^2+y^2+z^2}}\right)$

$=\left(\dfrac{q}{4\pi\epsilon_o}.\dfrac{-x}{(x^2+y^2+z^2)^{\frac{3}{2}}}\right)$

$= -\dfrac{q}{4\pi\epsilon_o}.\dfrac{x}{r^3}$

Similarly, $\dfrac{\partial V}{\partial y} = \left(\dfrac{q}{4\pi\epsilon_o}.\dfrac{-y}{(x^2+y^2+z^2)^{\frac{3}{2}}}\right)$

$= -\dfrac{q}{4\pi\epsilon_o}.\dfrac{y}{r^3}$

And,

$\dfrac{\partial V}{\partial z} = \left(\dfrac{q}{4\pi\epsilon_o}.\dfrac{-z}{(x^2+y^2+z^2)^{\frac{3}{2}}}\right)$

$= -\dfrac{q}{4\pi\epsilon_o}.\dfrac{z}{r^3}$

Putting these values, $grad\ V = -\frac{q}{4\pi\epsilon_o}.\frac{x\hat{\imath}+y\hat{\jmath}+z\hat{k}}{r^3}$

$$grad\ V = -\frac{q}{4\pi\epsilon_o}.\frac{\vec{r}}{r^3}$$

$$grad\ V = -\frac{q}{4\pi\epsilon_o}.\frac{r.\hat{r}}{r^3}$$

$$grad\ V = -\frac{q}{4\pi\epsilon_o}.\frac{\hat{r}}{r^2}$$

$$-grad\ V = \frac{q}{4\pi\epsilon_o}.\frac{\hat{r}}{r^2}$$

$$E = \frac{q}{4\pi\epsilon_o}.\frac{\hat{r}}{r^2}$$

Electric Current

Thursday, November 20, 2025 5:11 PM

The time rate of flow of charge through a conductor is called current. If a charge "Q" flows through any cross-section of a conductor in time "t", then the current I is given by:

$$I = \frac{Q}{t}$$

The SI unit of current is Ampere, which can be defined as, “when one coulomb charge flows through a cross-section in one second, then the current flowing is one ampere”.

Types of Current

Conventional Current

Conventional Current is the flow of positive charge from the positive terminal to the negative terminal of a battery

Electric Current

Electric Current is the actual flow of electrons, which move from negative terminal to the positive terminal.

Current Density

Thursday, November 20, 2025 5:16 PM

The current flowing per unit area is called the current density. It is a vector quantity and the SI unit of this quantity is Ampere per square meter ($\frac{A}{m^2}$).

$$J = \frac{I}{A}$$

Where,
J=Current Density
I=Current
A=Cross Sectional Area

Drift Velocity

The average velocity with which charge carriers move under the effect of electric current.
Distance travelled by the charge in time

$$t = \frac{1}{v_d}$$

Volume of wire through which the charge moves

$$V = A.v_d.t$$

$$Total\ Charge = Q = n.e.(A.v_d.t)$$

Where,
e=charge of each carriers
n=no. of electrons per unit volume

So, $I = \frac{Q}{t}$

$$I = \frac{n.e.(A.v_d.t)}{t}$$

$$I = n.e.(A.v_d)$$

$$Then, J = \frac{I}{A}$$

$$J = \frac{n.e.(A.v_d)}{A}$$

$$J = n.e.v_d$$

We follow our convention for positive current density which means we must take the direction of J opposite to that of v_d.

$$J = -n.e.v_d$$

Resistivity and Conductivity

Thursday, November 20, 2025 5:24 PM

Resistance

Resistance (R) is the measure of how much a specific object opposes the flow of electric current. Resistance is the ratio of voltage to current.

$$R = \frac{V}{I}$$

Its unit is Ohm.

Consider a conductor of length L and cross-sectional area A as shown in the figure. It is found experimentally that the resistance of conductor is directly proportional to length L of conductor and inversely proportional to cross-sectional area A:

$$R \propto L$$

$$R \propto \frac{1}{A}$$

Combining both

$$R \propto \frac{L}{A}$$

$$R = \rho \frac{L}{A}$$

Resistivity

The resistance of a meter cube of a substance is called resistivity or specific resistance. It is denoted by the symbol ρ. Its SI unit is ohm-meter.

$$R = \rho \frac{L}{A}$$

Where ρ is constant of proportionality, called resistivity or specific resistance of conductor.

$$\rho = \frac{RA}{L}$$

This is the expression of resistivity for a conductor having length L and cross-sectional area A.

Conductance

The **conductance** of a meter cube of a substance is called conductivity (or specific conductance). It is denoted by the symbol σ(sigma). Its SI unit is **Siemens per meter** ($^{S}/_{m}$).

Consider the same conductor of length L and cross-sectional area A. It is found experimentally that the **Conductance (G)** of the conductor is directly proportional to the Area (A) and inversely proportional to the Length (L), this is the exact opposite of resistance.

$$G \propto A$$

$$G \propto \frac{1}{L}$$

Combining both,

$$G \propto \frac{1}{L}$$

$$G = \sigma \frac{A}{L}$$

Where, σ the constant of proportionality, called conductivity of conductor

$$\sigma = G \frac{A}{L}$$

Ohms's Law

Thursday, November 20, 2025 5:29 PM

It states that:
“The current flowing through a conductor is directly proportional to the applied potential difference if all physical states remain same.”
If "V" is the potential difference between the ends of conductor and "I" is the current flowing through it, then the Ohm's law is described mathematically as:
$V \propto I$
$V = RI$
Where R is the constant of proportionality, called resistance of the conductor. It is described as the opposition offered by conductor to the flow of current. In system international, its unit is ohm. It is a macroscopic quantity. Its corresponding microscopic quantity is resistivity.

Application of Ohm's Law

Thursday, November 20, 2025 5:38 PM

Applications of Ohm's Law

1. **Circuit Calculations:** Helps find any one quantity (V, I, or R) if the other two are known.
2. **Designing Electronic Circuits:** Used to decide correct resistor values for TVs, chargers, and LEDs.
3. **Selecting Proper Fuse:** Helps calculate maximum current to select the right fuse rating for safety.
4. **Battery Calculations:** Used to check how much current a battery will supply to a device.
5. **Household Wiring:** Electricians use it to choose correct wire thickness and load capacity for safe wiring

Hall Effect

Thursday, November 20, 2025 5:39 PM

Corrected Notes: The Hall Effect

1. Setup & Definition

When a conductor carrying current is placed in a transverse magnetic field (perpendicular to current), a potential difference appears across its width. This is the Hall Voltage (V_h).

- **Geometry:** A strip of width w and thickness t.
- **Fields:** Magnetic Field (B) is perpendicular to the surface. Current (I) flows along the length.

2. Forces at Equilibrium

The charge carriers (electrons) experience the Lorentz Force:

$F = q(E + v \times B)$

At equilibrium, the magnetic force deflecting the electrons is balanced by the electric force (Hall field) created by charge separation. The net force is zero (F=0).

$F_e + F_m = 0$

$qE + q(v_d \times B) = 0$

$E = -(v_d \times B)$

Magnitude Condition:

Since the forces balance, their magnitudes are equal:

$qE = qv_dB$

$E = v_dB$

(Equation 1)

3. Deriving Hall Voltage (V_h)

The Hall Voltage is the electric field times the width (w) of the conductor:

$V_h = E.w$

Substitute E from Equation 1:

$V_h = v_dBw$

(Equation 2)

4. Substituting Current (I)

- Formula for Current: $I = nqAv_d$
- For electrons: q = e, and $Area\ A = width \times thickness = \text{w}.t$
- $I = ne\ (w.t)\ v_d$

Rearrange to solve for drift velocity (v_d):

$$v_d = \frac{I}{newt}$$

5. Final Formula

Substitute this v_d back into Equation 2:

$$v_h = \left(\frac{I}{newt}\right)Bw$$

The width (w) cancels out:

$$v_h = \frac{IB}{net}$$

(This is the standard formula for Hall Voltage)

6. The Hall Coefficient (R_h)

The Hall Coefficient relates the electric field to the current density.

- **Definition:** $R_h = \frac{E}{JB}$
- **Current Density (J):** $J = \frac{I}{A} = \frac{I}{wt}$
- **Electric Field (E):** $E = \frac{V_h}{w}$

Substituting these into the definition:

$$R_h = \frac{V_h/_w}{\left(I/_{wt}\right)B} = \frac{V_h . t}{I . B}$$

Now, substitute the V_h formula derived above ($V_h = \frac{IB}{net}$):

$$R_h = \frac{\frac{IB}{net} . t}{I . B}$$

Everything cancels except n and e:

$$R_h = \frac{1}{ne}$$

(Technically $R_h = \frac{-1}{ne}$ for electrons due to negative charge, but magnitude is often written as $\frac{1}{ne}$)

Biot-Savart Law

Thursday, November 20, 2025 5:46 PM

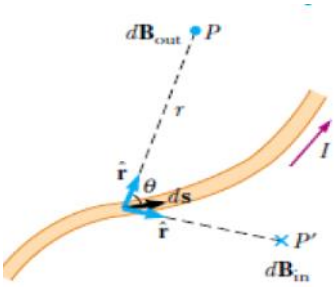
The experimental observations of
Biot and Savart about the magnetic field produced by a current carrying conductor at some point in space are as follows:

- The vector dB is perpendicular both to ds (which points in the direction of the current) and to the unit vector $\hat{r}$ directed from ds toward P.
- The magnitude of dB is inversely proportional to r^2, where r is the distance from ds to P.
- The magnitude of dB is proportional to the current and to the magnitude |ds| of the length element ds.
- The magnitude of dB is proportional to $\sin\theta$, where θ is the angle between the vectors ds and $\hat{r}$

These observations are summarized in the mathematical expression known today as the Biot-Savart law:

$|dB| \propto \frac{Ids\sin\theta}{r^2}$

$|dB| = \frac{\mu_o}{4\pi}\frac{Ids\sin\theta}{r^2}$



Where $\frac{\mu_o}{4\pi}$ is the constant of proportionality and $\mu_o = 4\pi \times 10^{-7}\frac{Wb}{A.m}$ is called the permeability of free space.

In vector form:

$|dB| = \frac{\mu_o}{4\pi}\frac{Ids\sin\theta}{r^2}\hat{n}$

Where $\hat{n}$ is the unit vector showing the direction of field and is determined by right hand rule.

If $\hat{I}$ is a unit vector along current element

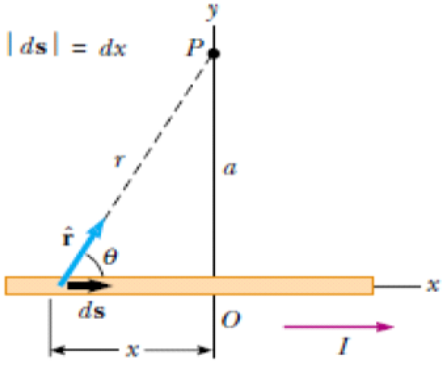
And $\hat{r}$ is the unit vector along position vector r of point P , then

$\hat{I} \times \hat{r} = |\hat{I}||\hat{r}|sin\theta\hat{n} = (1)(1)sin\theta\hat{n} = sin\theta\hat{n}$

The equation will become:

$dB = \frac{\mu_o}{4\pi}\frac{Ids\sin\theta}{r^2}\hat{n}$

$dB = \frac{\mu_o}{4\pi}\frac{Ids}{r^2}(\hat{I}\times\hat{r})$



As $\hat{I}$ is a unit vector along current element, therefore, $ds = ds\hat{I}$

$dB = \frac{\mu_o}{4\pi}\frac{I(ds\times\hat{r})}{r^2}$

As we know, $\hat{r} = \frac{r}{r}$

$dB = \frac{\mu_o}{4\pi}\frac{Ids}{r^2}\times\frac{r}{r}$

$dB = \frac{\mu_o}{4\pi}\frac{Ids\times r}{r^3}$

The magnetic induction B at point due to whole wire is obtained by integrating eq.

$B = \int dB = \frac{\mu_o}{4\pi}\int\frac{Ids\times r}{r^3}$

Force Between 2 Parallel Current Carrying Conductor

Sunday, November 23, 2025 6:14 PM

Consider two long, straight, parallel wires separated by a small distance a carrying currents I_1 and I_2 , respectively. The current passing through each wire produces a magnetic field around it
and each wire is placed in the magnetic field produced by the other. The wire 2, which carries a current I_2 creates a magnetic field B_2 at the location of wire 1, which can be found out by using the expression:

$$B_2 = \frac{\mu_o I_2}{2\pi a}$$

The direction of B_2 is perpendicular to wire 1. The magnetic force on wire 1 due to magentic fore of wire 2 is:

$$F_1 = I_1 L \times B_2$$

The magnitude of magnetic force on wire 1 is:

$$F_1 = I_1 L B_2 \sin 90^\circ$$

Because L and B_2 are perpendicular to each other

$$F_1 = I_1 L B_2$$

$$F_1 = I_1 L \left(\frac{\mu_o I_2}{2\pi a}\right)$$

$$F_1 = L \frac{\mu_o I_1 I_2}{2\pi a}$$

Where, L is the length of the conductor. The direction of force F_1 is towards wire 2.
Similarly, the force on wire 2, due to magnetic force of wire 1, carrying current I_2 is expressed as:

$$F_2 = L \frac{\mu_o I_1 I_2}{2\pi a}$$

The force F_2 is directed towards wire 1. The forces are equal and opposite, so, will attract each other. If the direction of element in one wire is opposite to that in the other, the wires will repel each other.

Ampere's Law

Sunday, November 23, 2025 6:10 PM

The ampere law is stated as,
"The line integral of magnetic induction B due to current I around any close loop is μ_o times the current enclosed"
$B = \mu_o I$

Integral Form of Ampere's Law

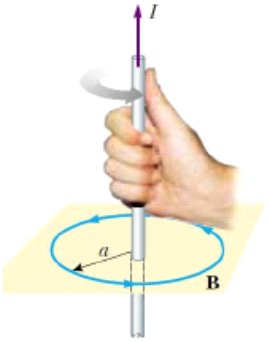
Consider a straight current carrying wire. The magnetic field will be produced around the conductor as the current flows through it. The magnetic field strength due to this current carrying wire at the distance a is expressed as:

$$B = \frac{\mu_o I}{2\pi R}\hat{k}$$

The symmetry shows that the magnetic induction is same everywhere on the circumference of circle. Thus, the expression can be expressed in general form as:

$$\oint B.dI = \oint \frac{\mu_o I}{2\pi a}\hat{k}.dI$$

$$= \frac{\mu_o I}{2\pi a}\oint \hat{k}.dI$$



As we know that $\hat{k}\ is\ paralell\ to\ dI$ so,
$\hat{k}.dI = \hat{k}dIcos90^{\circ}$
$\hat{k}.dI = (1)(dl)(1) = dl$

$$= \frac{\mu_o I}{2\pi a}\oint dl$$

$dl = ad\theta$

$$\oint B.dI = \frac{\mu_o I}{2\pi a}\oint ad\theta$$

$$= \frac{\mu_o Ia}{2\pi a}\oint d\theta$$

$$= \frac{\mu_o Ia}{2\pi a}(2\pi)$$

$$\oint B.dI = \mu_o I$$

This is called Ampere's Circuital Law.
As $I = \int J.ds$, therefore

$$\oint B.dI = \mu_o \int J.ds$$

This is integral form of Ampere's Law.

Differential Form of Ampere's Law

The Ampere's Law is expressed in integral form as:

$$\oint B.dI = \mu_o \int J.ds$$

Using Stoke's Theorem on L.H.S. we get,

$$\oint B.dI = \int_s curlB.ds$$

So, the equation becomes,

$$\int_s curlB.ds = \mu_o \int_s J.ds$$

$$\int_s curlB.ds - \mu_o \int_s J.ds = 0$$

$$\int_s (curlB - \mu_o J).ds = 0$$

$curlB - \mu_o J = 0$
$curlB = \mu_o J$
This is differential form of Ampere's Law

Magnetic Field Due to Solenoid

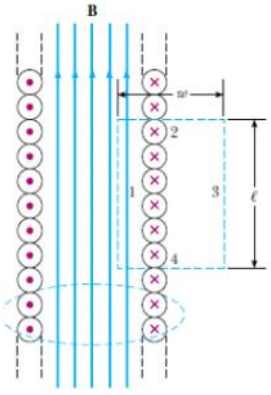
Sunday, November 23, 2025 8:32 PM

A solenoid is a cylindrical frame tightly wounded by an insulated wire. The magnetic field strength outside the solenoid is negligible as compared to the field inside it. We want to find out magnetic field intensity at point inside a current carrying solenoid.

For this we consider an Amperian loop. The symmetry shows that the close path can be divided into four elements1, 2, 3 and 4. By Ampere's law:

$$\oint B.dl = \mu_o \times (Current\ enclosed)$$

The integral $\oint B.dl$ can be written as the sum of four integrals



$$\oint_1 B.dl + \oint_2 B.dl + \oint_3 B.dl + \oint_4 B.dl = \mu_o \times (Current\ enclosed)$$

Here , $\oint_2 B.dl\ and\ \oint_3 B.dl = 0$ because the angle between B and dl is 90° also $\oint_3 B.dl = 0$ as the field outside solenoid is negligible

$$\oint_1 B.dl = \mu_o \times (Current\ enclosed)$$

$$\oint_1 Bdlcos0^\circ = \mu_o \times (Current\ enclosed)$$

$$\oint_1 Bdl = \mu_o \times (Current\ enclosed)$$

Since the magnetic field strength for the loop element 1 is constant inside the solenoid. So,

$$B\oint_1 dl = \mu_o \times (Current\ enclosed)$$

$$Bl = \mu_o \times (Current\ enclosed)$$

where is the length of element 1 of Amperian loop.

To find out the current enclosed by the loop, we consider that there are n turns per unit length of a solenoid. Then

$$Number\ of\ turns\ in\ length\ l\ of\ solenoid = nl$$

If I is the current flowing through solenoid, then

$$Current\ enclosed\ by\ the\ Amperian\ Loop = nlI$$

Therefore,

$$Bl = \mu_o \times nlI$$

$$Bl = \mu_o nI$$

This is the expression of magnetic field strength due to a current carrying solenoid.

Magnetic Field Due to Toroid

Sunday, November 23, 2025 8:46 PM

A toroid consists of a conducting wire wrapped around a ring (a torus) made of a nonconducting material.
We want to find out magnetic field strength at any point inside a toroid. For this we consider circular Amperian loop of radius. By symmetry, we see that the magnitude of the field is constant on this circle and tangent to it, therefore:
$B.ds = Bds$
Where, ds is small element of amperian loop.
By applying Ampere's Law:

$$\oint B.ds = \mu_o \times (Current\ Enclosed)$$

$$\oint Bds = \mu_o \times (Current\ Enclosed)$$

$$B\oint ds = \mu_o \times (Current\ Enclosed)$$

As, $\oint ds = 2\pi r$
$B \times 2\pi r = \mu_o \times (Current\ Enclosed)$
If N are the number of turns of the toroid and I is the current in the toroid, then
$Current\ enclosed\ by\ the\ Amperian\ Loop = NI$
$B \times 2\pi r = \mu_o \times NI$
$B = \frac{\mu_o NI}{2\pi r}$
This is the expression of magnetic field strength inside a current carrying toroid.

Faraday's Law of Electromagnetic Induction

Sunday, November 23, 2025 8:57 PM

The magnitude of induced emf in a circuit is directly proportional to the rate of change of magnetic flux.

$$\varepsilon = \frac{-d\varphi}{dt}$$

Differential form of Faraday's Law

We know, $Potential\ difference = V = \oint \vec{E}.\overrightarrow{dl}$

From Faraday's Law

$$V = \varepsilon = \frac{-d\varphi}{dt} = -\frac{-d}{dt}\int_s \vec{B}.\overrightarrow{ds}$$

$$= -\int_s \frac{\partial \vec{B}}{\partial t}.\overrightarrow{ds}$$

Comparing 2 Eq.

$$\oint \vec{E}.\overrightarrow{dl} = -\int_s \frac{\partial \vec{B}}{\partial t}.\overrightarrow{ds}$$

$$\int_s curl\vec{E}.\overrightarrow{ds} = -\int_s \frac{\partial \vec{B}}{\partial t}.\overrightarrow{ds}$$

$$\int_s curl\vec{E}.\overrightarrow{ds} + \int_s \frac{\partial \vec{B}}{\partial t}.\overrightarrow{ds} = 0$$

$$\int_s \left(curl\vec{E} + \frac{\partial \vec{B}}{\partial t}\right).\overrightarrow{ds} = 0$$

$$curl\vec{E} + \frac{\partial \vec{B}}{\partial t} = 0$$

$$curl\vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

This is differential form of Faraday's law of electromagnetism induction.

Integral form of Faraday's Law

$$\varphi_B = \int_s \vec{B}.\overrightarrow{dA}$$

EMF in terms of electric field

$$\varepsilon = \int_c E.dl$$

Emf by Faraday's observation

$$\varepsilon \propto \frac{-d\varphi_B}{dt}$$

From eq. 1

$$\varepsilon = \frac{-d}{dt}\int_s \vec{B}.\overrightarrow{dA}$$

Combing both equations, $\int_c E.dl = -\frac{d}{dt}\int_s B.dA$

This is the integral form of Faraday's law of electromagnetic induction.

The negative sign is due to the fact that the direction of induced current is such that it opposes the cause producing it.

Faraday's Experiment

Wednesday, December 3, 2025 9:04 PM

Experiment 1

Moving Magnet & Coil

Setup

Coil connected to galvanometer such that magnet can move toward & away from coil.

Observations

- Magnet moves then, Galvanometer deflects
- If magnet is stationary then there is no deflection on galvanometer

Conclusion

Induced emf occurred only when magnetic flux changes.

Experiment 2

Two Coil (Mutual Induction)

Setup

- Primary Coil with battery k switch
- Secondary Coil with galvanometer nearby

Observation

- Closing or Opening switch in primary coil causes galvanometer in secondary coil to deflect.
- Steady current gives no deflection

Conclusion

Changing current in one coil introduces emf in nearby coil (Mutual Induction).

Lenz's Law

Sunday, November 23, 2025 9:13 PM

Statement:
The direction of induced current is such that it opposes its own cause.
Explanation:
Consider a bar magnet moves toward a stationary metal loop, as in figure (a). As the magnet moves to the right toward the loop, the external magnetic flux through the loop increases with time.
As the result, the induced current set up in the loop which produces magnetic field, as illustrated in figure (b). Knowing that like magnetic poles repel each other, we conclude that the left face of the current loop acts like a north pole and that the right face acts like a south pole.
If the magnet moves to the left, as in figure (c), its flux through the area enclosed by the loop decreases in time. Now the induced current in the loop produces the magnetic field as shown in figure (d). In this case, the left face of the loop is a south pole and the right face is a north pole.

Motional EMF

Tuesday, November 25, 2025 7:08 PM

The emf induced in a loop by moving it a magnetic field is called motional emf. Consider a straight conductor of length is placed in a magnetic field which is directed into the plane of paper. For simplicity, we assume that the conductor is moving in a direction perpendicular to the field with constant velocity under the influence of some external agent.
As the bar is pulled to the right with a velocity under the influence of an applied force , free charges in the bar experience a magnetic force directed along the length of the bar. This force sets up an induced current because the charges are free to move in the closed conducting path. The magnetic flux passing through the circuit is:

$\varphi_b = B(Area) = B(l)(x)$

The induced emf can be determined using Faraday's law of electromagnetic induction:

$$\varepsilon = \frac{-d\varphi_b}{dt}$$

$$\varepsilon = \frac{-d(Blx)}{dt}$$

$$\varepsilon = -Bl\frac{dx}{dt}$$

$$\varepsilon = -Blv$$

This is the expression of induced emf in moving conductor. Due to this induced emf, the current induce in the conductor and is given by:

$$I = \frac{\varepsilon}{R}$$

$$I = \frac{-Blv}{R}$$

Where R is the resistance of the loop.
This induced current gives rise to the magnetic forces $F_B = I(l \times B)$ acting on the conductor. The magnitude of magnetic force will be:

$F_B = IlB$

Putting the values of I

$$F_B = \left(\frac{-Blv}{R}\right)lB$$

$$F_B = \frac{-B^2l^2v}{R}$$

The power expended in moving the conductor can be find out by the expression:

$P = F_{app}v$

$$P = \frac{-B^2l^2v}{R}v$$

$$P = \frac{-B^2l^2v^2}{R}$$

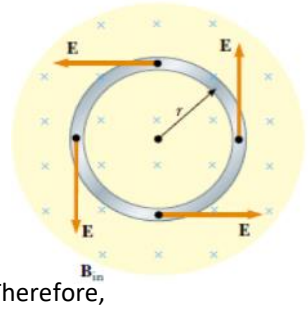
This is the expression of power delivered to the conductor t move in magnetic field.

Induced Electric Field

Sunday, November 23, 2025 9:15 PM

A straight current carrying conductor surrounded by a magnetic field. This magnetic field is surrounded by magnetic lines of force which are found to be concentric circles having their centers on the wire.
Similarly a changing magnetic field is surrounded by an electric field called induced electric field. This induced electric field is represented by electric lines of force which are found to be concentric circles.
We want to find out the expression of induced electric field due to changing magnetic field. For this, we consider a loop of conducting wire placed in a uniform magnetic field. This magnetic field may be applied by an electromagnet. By varying the current in the electromagnet, we can change the strength of magnetic field.
When B is changed, the magnetic flux through the loop
changes. So by using the Faraday's law and Lenz law we can find the magnitude and direction of induced emf and induced current.
Consider a circular path of radius r as shown in the figure below. Since emf is defined as the work done in moving a unit positive charge from one point to another point. For a close path, the work done per unit charge is expressed as:

$$\varepsilon = \oint E.dr$$



If E is parallel to dr

$$\varepsilon = \oint Edrcos0^{\circ}$$

$$\varepsilon = \oint Edr$$

If the rate of change of magnetic flux is constant, then the induced electric field will also be constant. Therefore,

$$\varepsilon = E\oint dr$$

$$\varepsilon = E(length\ of\ closed\ path)$$
$$\varepsilon = E(2\pi r)$$
By Faraday's law of electromagnetic induction, the induced emf in a coil of single loop will be:

$$\varepsilon = \frac{d\varphi_b}{dt}$$
Comparing the equations,

$$E(2\pi r) = \frac{d\varphi_b}{dt}$$

$$E = \frac{1}{2\pi r}\frac{d\varphi_b}{dt}$$
This is the expression of magnitude of induced electric field.

Basic Equations of Electromagnetism

Tuesday, November 25, 2025 7:35 PM

Topic: Maxwell's Equations

1. Overview

- **Origin:** Proposed by Scottish scientist James Clerk Maxwell.
- **Purpose:** These equations describe the fundamental laws of **electromagnetism**, which deals with the interaction between electric and magnetic fields.
- **Electromagnetic Field:** Formed by the combination of electric and magnetic fields.

2. Maxwell's First Equation: Gauss's Law for Electricity

- **Statement:** The total electric flux through a closed surface is proportional to the total charge enclosed by that surface.
  Mathematical Form:
  $\oint E \cdot dA = \frac{Q}{\epsilon_o}$
  (Where ϵ_o is the permittivity of free space, the capability of vacuum to permit electric fields).

Physical Interpretation:

- Electric flux is the integral of the electric field over the surface area.
- **Source & Sink:** Positive charges act as sources (lines outward) and negative charges act as sinks (lines inward).
- **Result:** Electric monopoles (isolated charges) **exist**.

3. Maxwell's Second Equation: Gauss's Law for Magnetism

- **Statement:** The total magnetic flux through any closed surface is always **zero**.
- Mathematical Form:
  $\oint B \cdot dA = 0$
  Or in differential form: $\nabla \cdot B = 0$.

Physical Interpretation:

- Magnetic sources are always **dipoles** (North and South poles).
- Field lines leaving the North pole must eventually return to the South pole, forming **closed loops**.
- **Result:** Isolated **Magnetic Monopoles do NOT exist**.

4. Maxwell's Third Equation: Faraday's Law of Induction

- **Statement:** A changing magnetic flux through a surface induces an electromotive force (EMF) or voltage in a loop of wire.
  Mathematical Form:
  $\oint E \cdot dl = -\frac{d\phi_B}{dt}$
  Or in differential form: $\nabla \times E = -\frac{\partial B}{\partial t}$
  Physical Interpretation: A time-varying magnetic field produces an electric field around its boundary.

5. Maxwell's Fourth Equation: Ampere-Maxwell Law

- **Original Ampere's Law:** A current flowing through a wire induces a magnetic field around it. Direction is determined by the **Right Hand Rule**.
  $\nabla \times B = \mu_o J$

The Problem (Maxwell's Find):

- Ampere's law works for steady current but fails for non-steady current, such as charging a capacitor.
- **Mathematical Contradiction:** Taking the divergence of both sides: The divergence of a curl is always zero ($\nabla \cdot (\nabla \times B) = 0$), but for changing charge, the divergence of current is non-zero ($\nabla \cdot$

$J! = 0$). This creates a mathematical impossibility (0=non-zero).

- **Physical Contradiction:** If you draw a surface passing through the wire, you calculate a magnetic field. If you draw a "balloon" surface passing *between* the capacitor plates (where no wire exists), Ampere's law wrongly predicts zero magnetic field.

Maxwell's Correction (Displacement Current):

- Maxwell realized that the changing electric field between the capacitor plates acts like a current.
- As charge accumulates on the plates, the electric field increases steadily, producing a magnetic field.
  He introduced the **Displacement Current** term:
  $I_d = \epsilon_o \frac{d\phi_e}{dt}$
- **Final Corrected Equation:**
  $\oint B \cdot ds = \mu_o I + \mu_o \epsilon_o \frac{d\phi_e}{dt}$

Numericals

Wednesday, November 26, 2025 1:07 PM

Resistivity and Resistance

1. A rectangular bar of iron is $2.0\ cm$ by $2.0\ cm$ in cross section and $40\ cm$ long. Calculate its **resistance**, If the resistivity of iron is $11 \times 10^{-8}\Omega m$.

Given:

- $L = 40cm = 0.4m$
- $A = 2.0cm \times 2.0cm = 4.0cm^2 = 4 \times 10^{-4}m^2$
- $\rho = 11 \times 10^{-8}\Omega m$

To Find:

$R = ?$

Formula:

$$R = \rho\frac{L}{A}$$

Solution:

$$R = \frac{11 \times 10^{-8} \times 0.4}{4 \times 10^{-4}}$$

$$R = 1.1 \times 10^{-4}\Omega$$

Electric Current

2. The current in the electron beam of a typical video display terminal is $200\mu A$. **How many electrons strike the screen each minute**?

Given:

- $i = 200 \times 10^{-6}A$
- $t = 60s$
- $e = 1.6 \times 10^{-19}C$

To Find:

$n = ?$

Formula:

- $n = \frac{q}{e}$
- $q = it$

Solution:

$q = 200 \times 10^{-6} \times 60$

$q = 1.2 \times 10^{-2}C$

$$n = \frac{1.2 \times 10^{-2}}{1.6 \times 10^{-19}}$$

$n = 7.5 \times 10^{16} electrons$

Ohm's Law and Power

3. A heating wire has a resistance 72Ω and it is also connected to a battery of $120V$. Under what condition will it **dissipate more heat**?

(a) Its length is connected across a battery.

(b) The wire is cut in half and two halves are connected in parallel across battery.

Given:

- $V = 120V$
- $R = 72\Omega$

To Find:

$\mathbf{P} = ?$

Formula:

- $P = \frac{V^2}{R}$
- *Since wire is cut in 2 halves,*

$P = \frac{V^2}{\frac{R}{2}}$

$P = \frac{2V^2}{R}$

Solution:

$(\mathbf{a})\ P = \frac{(120)^2}{72}$

$P = 200W$

$(\mathbf{b})\ P = \frac{2(120)^2}{72}$

$P = 400W$

Since, wire is split in 2 halves

Total power=400+400=800W

The wire dissipates more heat in **condition (b)** (parallel connection), where the total power is $\mathbf{800W}$, which is **4 times** the power of the single wire

Forces in Magnetic/Electric Fields

4. An electric field of $1.5\ kV/m$ and a magnetic field of $0.44T$ act on a moving electron to produce no force. Find **minimum speed of electron**.

Given:

- $E = 1.5 \times 10^3\ kV/m$
- $B = 0.44T$

To Find:

$v = ?$

Formula:

$v = \frac{E}{B}$

Solution:

$v = \frac{1.5 \times 10^3}{0.44}$

$v = 3400ms^{-1}$

Magnetic Field of a Solenoid

5. A solenoid is $1.33m$ long and diameter $2.6cm$ carries as current of $17.8A$. The magnetic field inside the solenoid is $22.4mT$. Find the **total number of turns of solenoid**.Magnetic Field of a Solenoid

Given:

- $l = 1.33m$
- $i = 17.8A$
- $B = 22.4mT = 22.4 \times 10^{-3}T$
- $\mu_o = 4\pi \times 10^{-7}\ Tm/A$

To Find:

$N = ?$

Formula:

$N = \frac{Bl}{\mu_o I}$

Solution:

$N = \frac{22.4 \times 10^{-3} \times 1.33}{4\pi \times 10^{-7} \times 17.8}$

$N = 1332$

Magnetic Field due to a Long Straight Wire

6. Find value of **magnetic field** at a distance $2cm$ from a long straight conductor in which current $2mA$ flows.

Given:

- $R = 2 \times 10^{-2} m$
- $i = 2 \times 10^{-3} A$
- $\mu_o = 4\pi \times 10^{-7}\ Tm/A$

To Find:

$B = ?$

Formula:

$$B = \frac{\mu_o i}{2\pi R}$$

Solution:

$$B = \frac{4\pi \times 10^{-7} \times 2 \times 10^{-3}}{2 \times 3.14 \times 2 \times 10^{-2}}$$

$$B = 2 \times 10^{-8} T$$

Motional EMF

7. An automobile having radio antenna $110cm$ long travels at $90\ km/h$ in a region where earth magnetic field is $55\mu T$. Find the **maximum value of induced emf**.

Given:

- $L = 110cm = 1.10m$
- $B = 55\mu T = 55 \times 10^{-6} T$
- $v = 90\ km/h = \frac{90 \times 1000}{60 \times 60} = 25 ms^{-1}$

To Find:

$\varepsilon = ?$

Formula:

$\varepsilon = vBL$

Solution:

$\varepsilon = 25 \times 55 \times 10^{-6} \times 1.10$

$\varepsilon = 1.5 \times 10^{-3} V$

Magnetic Force on a Charge

8. A proton travelling at 23° w.r.t. magnetic field of strength $2.63mT$ experiences a magnetic force of $6.48 \times 10^{-17} N$. Find the **speed of proton**.

Given:

- $\theta = 23^\circ$
- $B = 2.63mT = 2.63 \times 10^{-3} T$
- $F = 6.48 \times 10^{-17} N$
- $q = 1.6 \times 10^{-19} C$

To Find:

$v = ?$

Formula:

$\vec{F} = q(\vec{v} \times \vec{B})$

$F = qvbsin\theta$

$$v = \frac{F}{qBsin\theta}$$

Solution:

$$v = \frac{6.48 \times 10^{-17}}{1.6 \times 10^{-19} \times 2.63 \times 10^{-3} \times \sin 23^\circ}$$

$v = 3.94 \times 10^{5} ms^{-1}$

Inductance

9. The current amplitude in a pure conductor in a radio receiver is to be $250\mu A$ when the voltage amplitude is $3.6V$ at a frequency of $1.6MHz$ (at the upper end of AM broadcast band). What **inductance reactance** is needed? What is **inductance**?

Given:

- $I = 250\mu A = 250 \times 10^{-6} A$
- $V = 3.6V$
- $f = 1.6MHZ = 1.6 \times 10^{6} Hz$

To Find:

- $X_L =?$
- $L =?$

Formula:

- $X_L = \dfrac{V}{I}$
- $L = \dfrac{X_L}{\omega}$

  $L = \dfrac{X_L}{2\pi f}$

Solution:

$X_L = \dfrac{3.6}{250 \times 10^{-6}}$

$X_L = 1.44 \times 10^{4}\Omega$

$L = \dfrac{1.44 \times 10^{4}}{2 \times 3.14 \times 1.6 \times 10^{6}}$

$L = 1.43 \times 10^{-3} H$

Wave Properties

10. A sinusoidal electromagnetic wave of frequency $40MHz$ travels in free space in the x-direction. Determine the **wavelength** and **period of the wave**.

Given:

- $c = 3 \times 10^{8} T$
- $f = 40MHz = 40 \times 10^{6} Hz$

To Find:

- $\lambda =?$
- $T =?$

Formula:

- $\lambda = \dfrac{c}{f}$
- $T = \dfrac{1}{f}$

Solution:

$\lambda = \dfrac{3 \times 10^{8}}{40 \times 10^{6}}$

$\lambda = 7.5m$

$T = \dfrac{1}{40 \times 10^{6}}$

$T = 2.5 \times 10^{-8} s$

Speed of Light

11. The distance between Earth and Sun is $1.5 \times 10^{11} m$. **How long does it take for light to cover this distance**?

Given:

- $S = 1.5 \times 10^{11} m$
- $c = 3 \times 10^{8} ms^{-1}$

To Find:

$t =?$

Formula:

$t = \dfrac{S}{v}$

$t = \dfrac{S}{c}$

Solution:

$$t = \frac{1.5 \times 10^{11}}{3 \times 10^{8}}$$
$$t = 500s$$

Polarization of Light

Saturday, December 27, 2025 8:31 AM

Light waves normally vibrate in many directions at the same time.
When light is forced to vibrate in only one direction, it becomes polarized.
So, polarization means filtering the light so that it vibrates in only one plane.

Why does polarization happen?

Light is a transverse wave. This means it vibrates perpendicular to its direction of travel.
Because of this property, we can control its vibration direction.

Types of Polarization

1. **Plane (Linear) Polarization**
   Light vibrates in only one direction.
2. **Circular Polarization**
   The electric field rotates in a circular pattern.
3. **Elliptical Polarization**
   A combination where the vibration pattern is elliptica**l.**

Methods of Polarization

1. **By Reflection**
   When light reflects from a shiny surface like water or glass at a special angle (Brewster's angle), it becomes polarized.
2. **By Transmission Through Polaroid**
   Polaroid sheets have microscopic crystals that allow only one direction of vibration.
3. **By Scattering**
   When sunlight hits dust or air molecules, it gets scattered and becomes partially polarized (why the sky looks blue).

Real-Life Applications

- Polaroid sunglasses remove glare from roads and water.
- 3D movies use polarization to give depth.
- Photographers use polarized filters to reduce reflections.

Reflection & Refraction of Light

Monday, December 29, 2025 9:52 PM

Reflection is the bouncing back of light when it hits a smooth surface like a mirror.

Laws of Reflection

1. Angle of incidence = Angle of reflection
2. Incident ray, reflected ray, and normal lie in the same plan

Types of Reflection

1. **Regular reflection**

Regular reflection happens when light hits a smooth and shiny surface and all the reflected rays come out in one direction. This creates a clear image.

Examples:

- Mirrors
- Still water
- Shiny metal

2. **Diffuse Reflection**

Diffuse reflection happens when light hits a rough or uneven surface, and the reflected rays scatter in many different directions. This is why no clear image is formed.

Examples:

- Walls
- Paper
- Clothes
- Wood

Applications

- Mirrors
- Periscopes
- Car headlights
- Reflectors on roads

Refraction is the bending of light as it passes from one transparent medium to another (e.g., from air to water). This happens because the speed of light changes when it enters a medium with a different density.

Laws of Refraction

1. **The Plane:** The incident ray, the refracted ray, and the normal at the point of incidence all lie in the same plane.
2. **Snell's Law:** The ratio of the sine of the angle of incidence (i) to the sine of the angle of refraction (r) is constant for a given pair of media.

   $n = \frac{\sin i}{\sin r}$

   (Where n is the refractive index)

Refractive Index (n) It is a measure of how much a medium slows down light.

$n = \frac{c}{v}$

- c = Speed of light in vacuum (3×108m/s)
- v = Speed of light in the medium

Cases of Refraction

1. **Rarer to Denser Medium (e.g., Air to Glass)**
    - Speed of light **decreases**.
    - The light ray bends **towards the normal**.
    - Angle of refraction is smaller than angle of incidence (r<i).
2. **Denser to Rarer Medium (e.g., Glass to Air)**
    - Speed of light **increases**.
    - The light ray bends **away from the normal**.
    - Angle of refraction is larger than angle of incidence (r>i).

Effects & Examples

- **Bent Pencil**
  A pencil placed in a glass of water appears bent at the surface due to the bending of light rays coming from the water.
- **Apparent Depth**
  The bottom of a swimming pool appears raised (shallower) than it actually is.
- **Twinkling of Stars**
  Caused by the refraction of starlight as it passes through different layers of the Earth's atmosphere.

Applications

- **Lenses**
  Convex and concave lenses (used in glasses, cameras, and microscopes) work on the principle of refraction.
- **Prisms**
  Used to split white light into a spectrum (dispersion).
- **Optical Instruments**
  Telescopes and binoculars.

Total Internal Reflection

Monday, December 29, 2025 10:08 PM

This is a phenomenon where light is completely reflected back into a medium instead of refracting out of it. It is the principle behind **Fiber Optics.**

- **The Scenario:** Light must travel from a **Denser Medium** (like water or glass, higher n_1) toward a **Rarer Medium** (like air, lower n_2).
- **The Mechanism:**
  1. When light moves Dense → Rare, it bends **away** from the normal.
  2. As you increase the **Angle of Incidence (θ_i)**, the **Angle of Refraction (θ_r)** also increases.
  3. **Critical Angle (θ_c):** There comes a specific angle of incidence where the refracted ray bends so much it skims perfectly along the boundary surface ($\theta_r = 90^\circ$).
  4. **TIR:** If you increase the angle *beyond* the critical angle ($\theta_i > \theta_c$), the light cannot escape. It reflects back inside entirely, acting like a perfect mirror.
- **The Formula (Snell's Law at Critical Angle):**
  $n_1 \sin\theta_c = 90^\circ$
  Since, $\sin 90^\circ = 1$
  $\sin\theta_c = \frac{n_2}{n_1}$

Two Source Interference

Monday, December 29, 2025 10:17 PM

This is the theoretical foundation for all interference patterns. It describes what happens when waves from two different sources overlap in space.

- **Superposition Principle:** When two waves meet at a point, the total displacement is the vector sum of their individual displacements ($y_{total} = y_1 + y_2$).
- **The Requirement (Coherence):** To see a steady interference pattern, the two sources must be Coherent. This means they must maintain a constant phase difference (they usually must have the exact same frequency).
- **Constructive Interference (Bright/Loud):**
  - The crest of one wave meets the crest of the other.
  - **Path Difference (ΔL):** The distance one wave travels minus the distance the other travels must be a whole number of wavelengths.
  - **Condition:** $\Delta L = m\lambda$ (where m=0,1,2...)
- **Destructive Interference (Dark/Quiet):**
  - The crest of one wave meets the trough of the other. They cancel out.
  - **Path Difference (ΔL):** The path difference must be a "half" wavelength.
  - **Condition:** $\Delta L = (m + 0.5)\lambda$

Young's Double Slit Interference

Monday, December 29, 2025 10:20 PM

This experiment is one of the most famous in physics because it provided the first clear proof that light behaves as a **wave**.

- **The Setup:** A single monochromatic light source passes through two tiny parallel slits (S_1 and S_2) separated by a distance d. A screen is placed at a distance D.
- **The Geometry:** Light from the two slits travels to a point P on the screen. Because the slits are separated, light from one slit travels a slightly longer distance than the other. This extra distance is the **Path Difference** ($\boldsymbol{\Delta L \approx dsin\theta}$).
- **Equations for Fringes:**
  - **Bright Fringes (Maxima):** Occur where path difference is $m\lambda$.
    $dsin\theta = m\lambda$
  - **Dark Fringes (Minima):** Occur where path difference is (m+0.5)λ.
    $dsin\theta = (m + \frac{1}{2})\lambda$
- **Linear Position on Screen (y):** Since the angle θ is usually very small, we can say $sin\theta \approx tan\theta = y/D$.
  - Position of m-th Bright Fringe:
    $y_m = \frac{m\lambda D}{d}$
- **Fringe Width (β):** The distance between two consecutive bright (or dark) fringes.
  $\beta = \frac{\lambda D}{d}$
  (Note: This tells us that fringes get wider if you use red light (higher λ) or move the screen further away (higher D).

Diffraction & Wave Theory

Wednesday, December 31, 2025 3:35 PM

Diffraction

Diffraction is the bending of waves around the corners of an obstacle or through an aperture (opening) into the region of geometrical shadow.

- **Condition:** Diffraction is most noticeable when the size of the obstacle or opening (a) is comparable to the wavelength of the wave (λ).
  $(a \approx \lambda)$.
- **Example:** Sound waves have long wavelengths (≈1m), so they easily diffract around doorways. Light waves have tiny wavelengths (≈500nm), so they usually travel in straight lines unless the opening is microscopic (like a slit).

Wave Theory (Huygens' Principle)

To explain diffraction, we use **Huygens' Principle**, which states:

- Every point on a wavefront acts as a source of secondary spherical wavelets.
- The new position of the wavefront is the tangent to these secondary wavelets.
- When a wavefront passes through a slit, the "bending" occurs because secondary wavelets at the edges spread out into the shadow region.
- **The Principle**
  Imagine every single point on a wavefront is a source of tiny, spherical "secondary wavelets" spreading forward.
- **The Result**
  When a flat wavefront hits a slit, the points inside the slit generate new spherical wavelets. These wavelets expand outward into the shadow region, causing the light to "bend" or flare out

Single Slit Diffraction

Wednesday, December 31, 2025 3:38 PM

1. The Setup

- A single slit of width **a** is illuminated by monochromatic light of wavelength λ.
- A screen is placed far away (distance D).

2. The Pattern

Instead of a single sharp line, you see a broad **Central Bright Maximum** flanked by alternating narrower dark and bright fringes (secondary maxima).

- **Central Maximum:** The brightest and widest part of the pattern (at θ=0).
- **Minima (Dark Fringes):** Occur where light from the top half of the slit cancels light from the bottom half.

3. Condition for Minima (Dark Fringes)

Unlike the double-slit experiment (where $dsin\theta = m\lambda$ was for *maxima*), in single-slit diffraction, the formula gives the **Dark Fringes**:

$asin\theta = m\lambda$

(Where $m = \pm1, \pm2, \pm3 \ldots$ and a is the slit width).

4. Width of Central Maximum

The central bright spot extends from the first dark fringe on the left ($m = -1$) to the first dark fringe on the right ($m = 1$).

- Angular width (2θ): Since $sin\theta \approx \theta = \lambda/a$, the total angular width is $2\lambda/a$.
- Linear width (y):
  $W = a2\lambda D$

5. Key Insight

If you make the slit narrower (decrease a), the central bright spot gets wider (diffraction increases).

Polarization of Electromagnetic Waves

Wednesday, December 31, 2025 3:38 PM

1. Nature of EM Waves

Electromagnetic waves are **Transverse Waves**. This means the Electric Field (E) and Magnetic Field (B) oscillate perpendicular to the direction of propagation.

2. Unpolarized vs. Polarized Light

- **Unpolarized Light:** The Electric field vectors vibrate in all random directions perpendicular to the wave's path (e.g., sunlight, bulb light).
- **Plane Polarized Light:** The Electric field vectors vibrate in **only one single plane** (e.g., light after passing through a polaroid filter).

Polarizing Sheets

Wednesday, December 31, 2025 3:44 PM

These are materials made of long-chain molecules aligned in a specific direction.

- **Transmission Axis:** The direction parallel to which the electric field can pass.
- **Absorption:** It absorbs any electric field component perpendicular to this axis.
- **Effect:** If unpolarized light (Intensity I_o) passes through a polarizing sheet, the intensity becomes half ($I_o/2$).